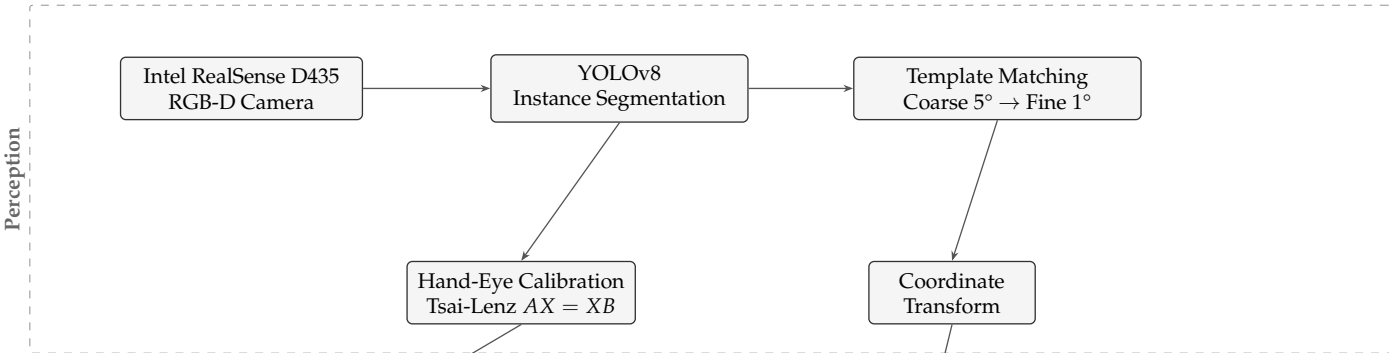
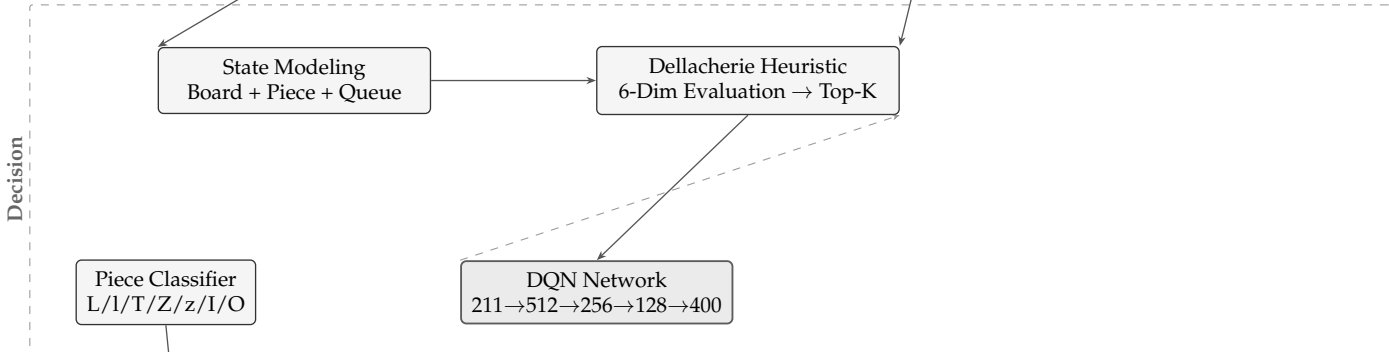


System Architecture: Perception → Decision → Execution

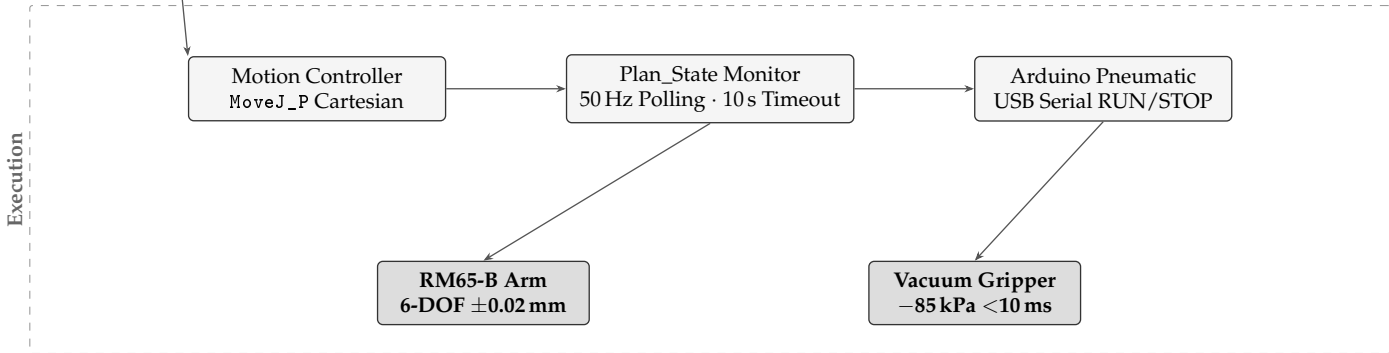
Perception Layer



Decision Layer



Execution Layer



ROS Node Communication Architecture

Hardware

RealSense D435
/camera/*

RM65-B Arm
/rm_driver/*

Arduino Uno
/dev/ttyUSB0

ROS Nodes (Python 3.8)

yolo_detector
Vision Pipeline

tetris_planner
Decision Engine

motion_controller
MoveJ_P Executor

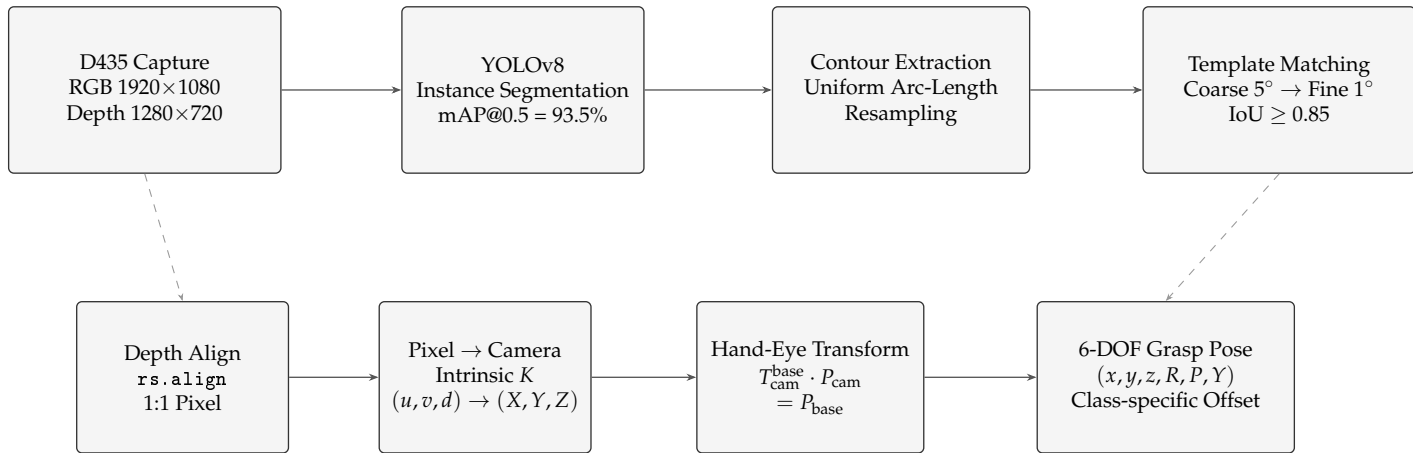
transform
TF Broadcaster

arduino_bridge
Serial Protocol

takephoto
Capture

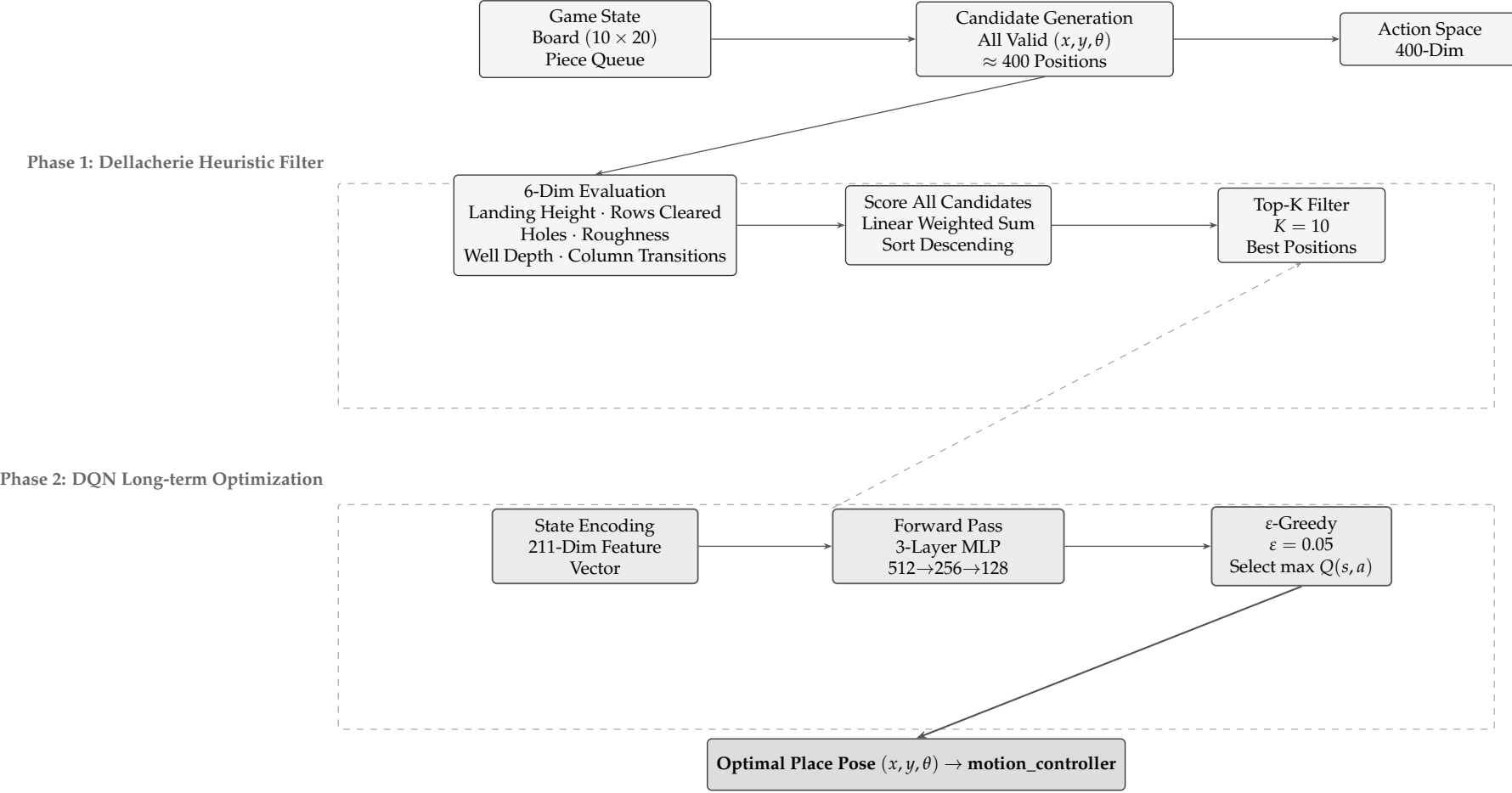
/camera/color/image_raw /camera/aligned_depth /vision/detections /planning/place_pose /rm_driver/MoveJ_P

Vision Pipeline: RGB-D Capture $\rightarrow$ 6-DOF Grasp Pose

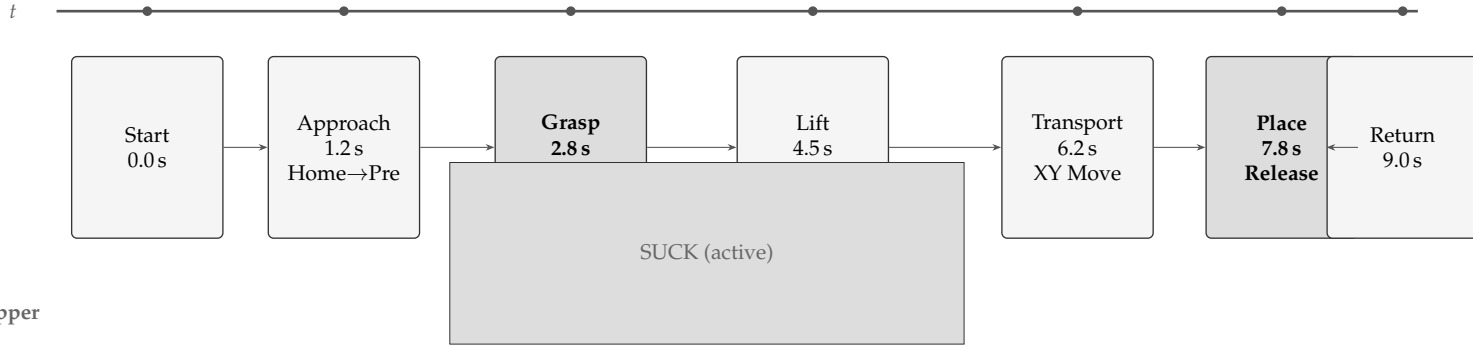


YOLOv8 mAP@0.5 = 93.5% | Template IoU ≥ 0.85 | Grasp Success = 96.2%

Hybrid Decision Strategy: Heuristic + Deep Reinforcement Learning



Pick-and-Place Motion Sequence (9.8 s Cycle)

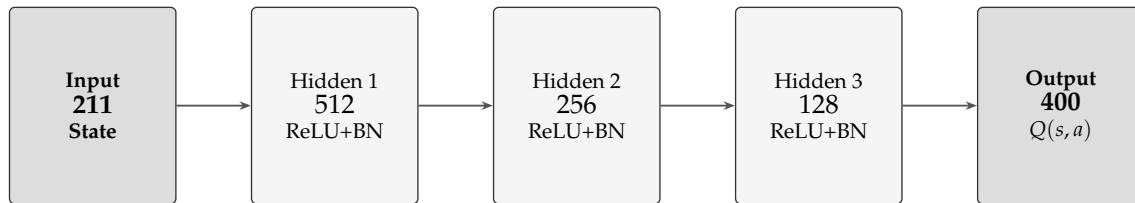


Plan_State 50 Hz Polling → wait_for_move() → Movement Complete → Timeout 10 s

Approach 1.2 s · Grasp 0.3 s · Lift 0.5 s · Transport 1.7 s · Place 0.5 s · Return 1.2 s = Total 9.8 s

Safety: Force Limit 50 N · Collision Detection · E-Stop < 10 ms

Deep Q-Network Architecture: 211-D State $\rightarrow$ 400-D Q-Values



Input Features (211-D):

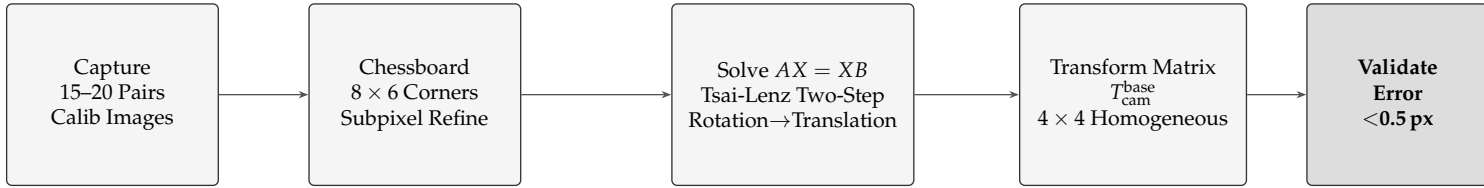
Board matrix: 200-D (10×20 grid, one-hot occupancy)
Piece queue: 7-D (current + next 6 pieces, class encoding)
Hole count: 1-D · Column max height: 1-D
Bumpiness sum: 1-D · Completed lines: 1-D

Training: Experience Replay Buffer (10K) · Target Network Sync (100 steps) · $\gamma = 0.99$ · $\eta = 1 \times 10^{-4}$ · Adam

Bellman Update: $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$

ϵ -greedy: ϵ decays from 1.0 $\rightarrow$ 0.05 over 10,000 episodes

Hand-Eye Calibration: Tsai-Lenz Algorithm



$$A \cdot X = X \cdot B$$

A : Robot end-effector pose change (forward kinematics)

B : Camera pose change (chessboard detection)

X : Unknown hand-eye transform (camera → end-effector)

Step 1: Rotation R_X via axis-angle + linear least squares

Step 2: Translation t_X from linear system with known rotation

Result: 6-DOF Transform Position $\pm 1.5 \text{ mm}$ Orientation $\pm 0.5^\circ$